

ELECTRIC VEHICLES CHARGING STATIONS IN KERALA

Project synopsis submitted in partial fulfillment for the
Award

CERTIFICATION

In

Electric Vehicle Course

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March Batch 2024

DATE: 25-03-2023

CHAPTER 1

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Abbreviations:

EV: Electric Vehicle

AC: Alternating Current

DC: Direct Current

PCS: Public Charging Stations

PPP: Private –Public Partnerships

EM: Electromagnetic

KSEB: Kerala State Electricity Board

KAL: Kerala Automobiles Ltd.

FAME: Faster Adoption and Manufacturing of Electric Vehicles in India

CPO: Charge Point Operator

CHAPTER 2

INTRODUCTION

The rise of electric vehicles in the automobile industry presents a transformative opportunity for India's transportation sector. Due to the large population and rapid economic growth, India needs to reduce its dependency on fossil fuels to reduce carbon emissions and adapt new sustainable alternatives. This case study explores the potential demand for EV charging stations infrastructures in Kerala as well as its difficulties in the context of Kerala. With a surge in privatization, liberalization and the expansion of distributed and renewable energy sources. And address challenges faced in EV adoption, economic risks, lack of charging infrastructure, risky maintenance of EV's, problems with its integration in the smart grid, range anxiety and auxiliary loads.

For faster adoption of EV's methods like public-private partnerships (PPP), which is a viable model for expanding up EV charging networks in across the country, And establishing more battery swapping stations and reduce the range anxiety and reduces the time delay for charging the batteries. Introduction and implementation of smart technologies like applications to locate charging stations contributes to the faster transitions to electric vehicles. Implementation of new technologies like drive and charge can accelerate the shift from conventional internal combustion vehicles to electric vehicles.

CHAPTER 3

CHARGING TECHNOLOGY

Charging Levels

There are two types of charging stations, one that supplies AC and the other that supplies DC. Battery always stores the energy as DC, and so even if AC is being received, a conversion takes place before it gets stored in the battery. So, Dc directly goes to the battery, while AC goes to the onboarding charging slot/converter and is then converted into DC for storage.

There are three levels of charging, L1, L2, and L3. L1 and L2 will supply AC and then, with the help of an onboarding converter, it converts that AC into DC. At the same time, L3 will directly supply DC to the EVs.

- L1 refers to the slow chargers that are used at home.
- L2 corresponds to the chargers used at stations available in public places.
- L3 corresponds to the charging received at the stations equipped with rapid chargers. They directly supply DC, they are the rapid chargers.

Different levels of charging in North America: SAE configurations	
Level-1 	• Supply from household outlet • Make use of EV's on-board charger • 120V 1ph AC; 12-16A • Charging power: 1.4KW or 1.9KW
Level-2 	• Supply from household outlet or EV Charge point • Make use of EV's on-board charger • 208-240V 1ph AC ; 12 to 80A (Typ. 30A) • Charging power: 2.5KW to 19.2KW (Typ. 7KW)
Level-3 	• Supply from 208-600V 3ph AC • Make use of off-board DC fast charger • 400A (Typ. 60A) • Charging power: up to 240KW (Typ. 50KW)

Figure 1: EV Charging Levels specifications

As per the information received from Ministry of Power, 12,146 number of public EV charging stations are operational across the country, as on 02.02.2024.

State wise operational public EV charging stations as on 02.02.2024

S.No.	State Name	No. of operational PCS
1	Andaman & Nicobar	3
2	Andhra Pradesh	327
3	Arunachal Pradesh	9
4	Assam	86
5	Bihar	124
6	Chandigarh	12
7	Chhattisgarh	149
8	D&D and DNH	1
9	Delhi	1886
10	Goa	113
11	Gujarat	476
12	Haryana	377
13	Himachal Pradesh	44
14	Jammu & Kashmir	47
15	Jharkhand	135
16	Karnataka	1041
17	Kerala	852
18	Lakshadweep	1
19	Madhya Pradesh	341
20	Maharashtra	3079
21	Manipur	17
22	Meghalaya	21
23	Nagaland	6
24	Odisha	198
25	Pondicherry	23
26	Punjab	158
27	Rajasthan	500
28	Sikkim	2
29	Tamil Nadu	643
30	Telangana	481
31	Tripura	18
32	Uttar Pradesh	582
33	Uttarakhand	76

34	West Bengal	318
	TOTAL	12,146

Types of charging

There are three ways of charging the battery of the EV

- Conductive charging
- Inductive charging
- Replacing the battery

Conductive charging uses direct contact between EV connector and charge inlet. The cable can be fed from standard electrical outlet or charging station. There are two methods employed in EV charging stations using conductive method. AC chargers and DC chargers.

Inductive charging uses electromagnetic (EM) field to transfer energy between two objects. Charging station is developed for this purpose. Energy is transmitted through inductive coupling to electrical device. This energy is used to charge batteries.

Battery swapping vs charging station



Figure 2: Difference between swapping station and charging stations

Battery Swapping

Battery swapping is a process where the depleted battery of an electric vehicle is easily replaced with a fully charged one that allows the driver to continue their journey without waiting for the vehicle to be charged. This method offers a quick and convenient solution, especially when compared to the time-consuming process of traditional charging. While battery swapping has gained attention for its efficiency, challenges such as standardization and infrastructure development need to be addressed for widespread acceptance.

Advantages

- Convenience
- Time efficiency
- Increased Range
- Flexibility

Charging Stations

An EV Charging Station is a facility designed to recharge the batteries of electric vehicles. It provides a dedicated space for users to connect their EVs and replenish their battery levels. These stations come in various forms, ranging from slow chargers for overnight charging to fast chargers that can recharge a significant portion of the battery within a short timeframe, typically 10 to 15 minutes for up to 80% capacity. EV charging stations play a crucial role in supporting the growing adoption of electric mobility by offering a convenient and efficient way to recharge electric vehicles.

Advantages

- Accessibility
- Versatility
- Cost effectiveness
- Compatibility

According to India Energy Storage Alliance (IESA), there are more than 20 active battery swapping operators, with around 1,000-1,100 swapping stations that hold more than 1.5 lakhs batteries, operating in the country.

Public Private Partnership (PPP)

Public-private partnerships are a viable model for scaling up EV charging networks in cities and also to provide reliable charging solutions to drive electrification in commercial passenger fleets. Public Private Partnership (PPP) means an arrangement between a Government and statutory entity / Government owned entity on one side and a private sector entity on the other, for the provision of public assets and/or public services, through investments being made and/or management being undertaken by the private sector entity.

Classification of charging infrastructure

1. Private charging refers to charging stations that are not available for public usage. They are dedicated to an individual apartment, office, or fleet. However, the fleet must have its land available for private charging. You can charge your electric vehicles the same way as you would at the public charging station. You can choose to operate by yourself or appoint a Charge Point Operator (CPO).

2. Public charging stations are open for all EVs. These charging stations can have multiple charging points so that many EV drivers can charge their vehicles at the same time.
3. Semi-public charging stations are dedicated to a set of people that are either residing at the place or visiting it. It is open to specific consumers although not owned by an individual.

EV Charging Stations Software Providers in India

Company Name	Application Name
Ather Energy	Ather Grid
TATA Power	TATA Power EZ charge
Battre	Batt:RE
Zeon charging	Zeon Charging
Statiq	Statiq
Magneta Charge Grid	Charge Grid
GOEC	GOEC
PlugShare	PlugShare
Fortum Charge & Drive	Charge & Drive

CHAPTER 4

CHARGING STATIONS IN KERALA

- Kerala has 852 public EV charging stations within the state as on 02.02.2024.
- Kerala has a wide variety of EV charging stations available. These include both public and private charging stations, as well as fast and slow chargers. Public charging stations are typically funded and

maintained by the government, while private charging stations are funded and maintained by private entities.

- The Kerala State electricity board (KSEB) operates all the public charging stations. In the private sector, GOEC is the largest operator of EV charging stations in Kerala.
- The cost of EV charging in Kerala depends on more than one factor. It includes the electricity rate, charging efficiency, and charging method.
 1. Electricity Rate – The most significant factor is the cost of electricity per unit (kWh). The base tariff of the Kerala State Electricity Board Stations is Rs 15/Unit, making it an affordable choice for all EV users.
 2. Battery Capacity – The capacity of the battery of an electric car is crucial in recognizing the charging cost and efficiency.
 3. Charging Efficiency – Electric vehicles need efficient charging. It requires fast conversion of electrical energy into usable energy for driving. The efficiency differs with the various EV models. The average efficiency range is around 80-90%.
- The number of charging is rapidly increasing due to significant investments by private companies in expanding their charging infrastructure.

CHAPTER 5

CHALLENGES

- Although EV charging infrastructure is rapidly expanding across Kerala, some challenges still need to be addressed. The first challenge is ensuring adequate availability of power supply at EV charging stations.
- The main reason for hindering the widespread adoption of electric vehicles in Kerala is the insufficient number of charging stations. Users need a reliable and accessible charging infrastructure to alleviate range anxiety and ensure the seamless operation of their EVs.
- The implementation of charging infrastructure requires significant investments and expenses. The high costs associated with installation, maintenance, and operation deter some stakeholders from establishing charging stations.
- The lack of standardized charging connectors and protocols presents a challenge for electric vehicle owners. Different manufacturers and regions often adopt varying standards, making it inconvenient for EV users to find compatible charging stations.

CHAPTER 6

GOVERNMENT INITIATIVES

- The government of Kerala has opened sub-stations to support charging infrastructure within the state.
- By placing charging stations near to the sub-stations is seen a promising starting point to address some of the challenges associated with EV adoption.
- The government of Kerala has implemented several initiatives to encourage EV adoption, including tax benefits, subsidies, and the installation of charging infrastructure across major cities and highways.
- Government of Kerala has accorded administrative sanction for the establishment of 32 EV charging stations to KSEB, through E-mobility Promotion fund. Accordingly, in the first phase, KSEBL established 6

Pilot EV Charging Stations in the corporation areas across the State with a view to build public awareness on E-Mobility.

- The Department of Heavy Industries, Government of India under FAME India Scheme Phase II initiated for deployment of EV charging Stations in select cities in India in August 2019. In Kerala, seven cities come under the scheme and are Trivandrum, Kollam, Malappuram, Kochi, Thrissur, Kozhikode and Kannur.
- Government is providing various incentives for manufacturers and supporting “vocal for local”. The key focus is on improving internal manufacturing ecosystem, thereby shifting from the import-dependent ecosystem to a consumption-driven economy. The manufacturers will enjoy various financial and regulatory benefits.
- Government is working to make an e-Auto manufacturing facility under Kerala Automobiles Ltd. (KAL), power distributors and energy system and storage providers.
- Kerala State Electricity Board has installed 63 DC fast charging stations for four-wheelers in all districts and 1166 pole-mounted charging stations for auto-rickshaws and two-wheelers in all constituencies in Kerala.
- The government has also prescribed the single-part EV tariff for public charging stations, this amount cannot exceed the Average Cost of Supply.
- The Kerala government is planning to implement drive and charge infrastructure for electric vehicles in the next financial year.

CHAPTER 7

CONCLUSION

Electric vehicle adoption is steadily increasing in Kerala due to proactive policy measures taken by the state government. An extensive network

of EV charging stations has been set up further to encourage EV adoption across various parts of the state. Kerala is among the states with highest number of electric vehicle charging stations in country ranked 4th behind Maharashtra, Delhi and Karnataka respectively. The Kerala State Electricity Board is also setting up many charging stations in the state to make the electric vehicle adoption quicker and efficient. The government is also promoting the adoption of electric vehicles by giving subsidies for setting up the charging infrastructure and implementing policies to support the shift. Also with implementation of new technologies like drive and charge and encouraging more public-private partnerships, Kerala is expected to rapidly develop in the electric vehicle charging infrastructure in the upcoming years.

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